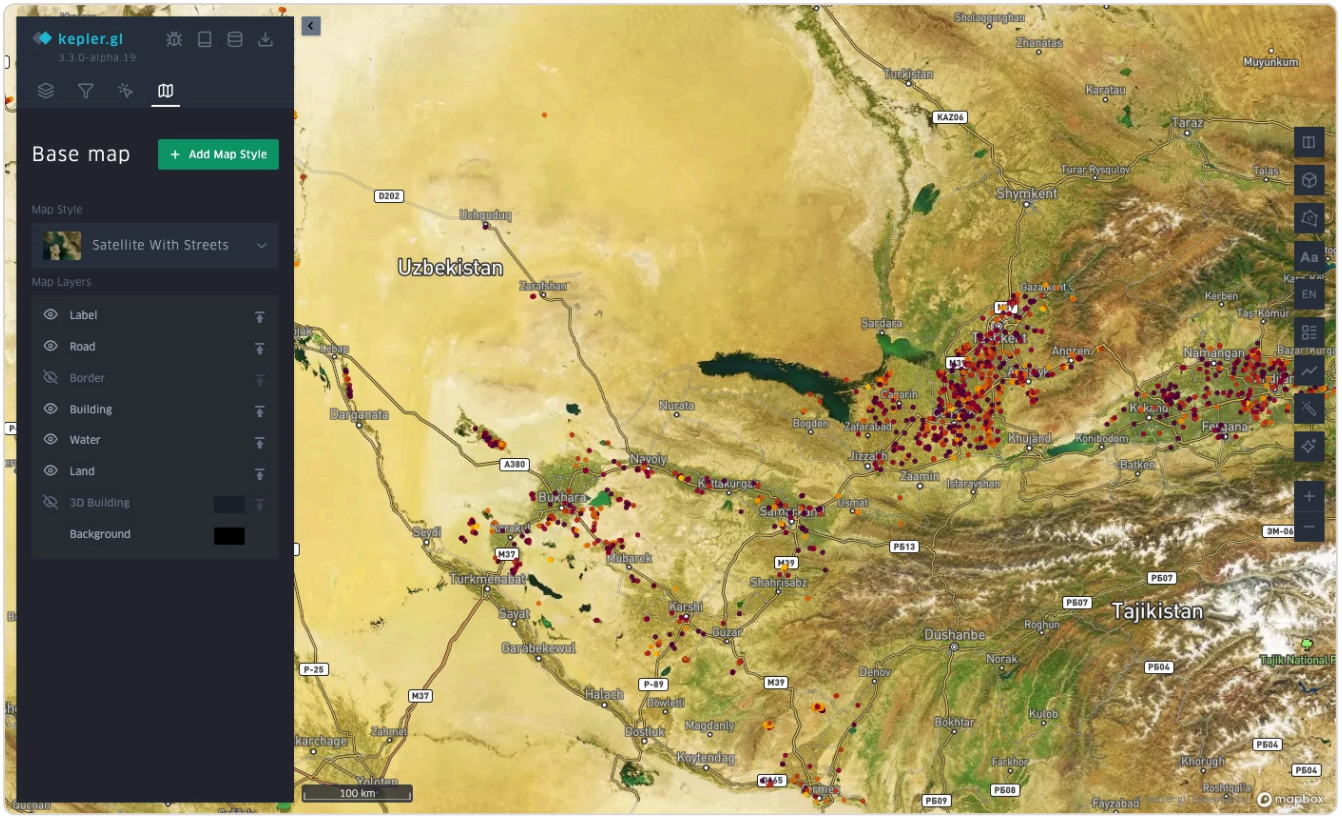
1-kun · Nuqtalar – Kosmosdan ko‘ringan yong‘inlar 🛰️🔥

Bu darsda siz NASA sun’iy yo‘ldoshlari 2025-yilda O‘zbekistonda aniqlagan barcha yong‘inlar xaritasini tuzasiz. **Hech narsa o‘rnatish shart emas, kod yozish ham shart emas.** Hamma bir xil ma’lumotdan boshlaydi, lekin har kim uni o‘zicha ko‘rsatadi. Shuning uchun guruhimizdagi har bir xarita boshqacha chiqadi.

⏱️ Vaqt	💻 Kerak bo‘ladi	🧰 Dastur
30–45 daqiqa	Chrome, Edge yoki Firefox o‘rnatilgan kompyuter	kepler.gl (bepul, brauzerda ishlaydi)



🔤 kepler.gl dasturi ingliz tilida. Shuning uchun bu darsda tugmalar nomi **inglizcha** yozilgan, ekranda qanday ko‘rsangiz, xuddi shunday.

🛰️ 0-qism – Ma’lumot bilan tanishing: NASA FIRMS yong‘inlari

Ma’lumot qayerdan keladi? NASAning ikkita sun’iy yo‘ldoshi, **Terra** va **Aqua**, Yer atrofida taxminan 700 km balandlikda aylanadi. Har birida kameraga o‘xshash **MODIS** degan asbob bor. MODIS oddiy suratga olishdan tashqari **issiqlikni** ham o‘lchaydi (issiqlik infraqizil nurlanishi).

Sun’iy yo‘ldosh yong‘inni qanday „ko‘radi“? MODIS yer yuzini taxminan **1 km × 1 km** kattalikdagi kvadratlarga bo‘ladi (ularni *piksel* deyiladi). Agar bitta kvadrat atrofidagilardan ancha issiq bo‘lsa, NASA kompyuterlari uni **issiq nuqta** (hotspot) deb belgilaydi. Har bir issiq nuqta faylimizda **bitta nuqta** bo‘ladi va o‘sha 1 km kvadratning markaziga qo‘yiladi.

> 💡 Demak, nuqta *„o‘sha paytda shu 1 km kvadratning qayeridadir nimadir yongan“* degani. Bu yong‘inning aniq joyi emas.

Sun’iy yo‘ldoshlar qachon kuzatadi? Ikkala sun’iy yo‘ldosh birgalikda O‘zbekiston ustidan kuniga taxminan **4 marta** o‘tadi: Terra soat 10:30 va 22:30 atrofida, Aqua soat 13:30 va 01:30 atrofida. Shuning uchun ma’lumotda **kunduzgi** ham, **tungi** ham kuzatuvlar bor.

Har bir issiq nuqta yong‘in emas. Neft va gaz konlaridagi gaz mash’allari va ba’zi zavodlar doim issiq bo‘ladi. NASA ularni *doimiy manbalar* deb ataydi. Bizning faylda ular **„Gas flare / industry“** (gaz mash’ali / sanoat) deb nomlangan.

Bu tizim FIRMS deb ataladi (Fire Information for Resource Management System). Undan butun dunyoda o‘t o‘chiruvchilar, fermerlar va olimlar har kuni foydalanadi.

Faylimizda nima bor

O‘zbekistonning 2025-yilgi ma’lumotlarini bitta jadval qilib tayyorladik: `fires_uzbekistan_2025.csv`. Har bir qator bitta nuqta (jami **4 649 ta nuqta**).

Ustun	Ma’nosi	Misol
`latitude`, `longitude`	Nuqta qayerda (kenglik, uzunlik)	41.25, 69.21
`date`	Aniqlangan kun	2025-03-14
`region`	Qaysi viloyatga tushadi	Surkhandarya
`fire_type`	O‘simlik yong‘ini yoki gaz mash’ali / sanoat	Vegetation fire
`day_night`	Kunduzi yoki kechasi ko‘ringan	Day
`season`	Qish / Bahor / Yoz / Kuz	Spring
`month`	Oy	03 Mar
`satellite`	Qaysi sun’iy yo‘ldosh ko‘rgan	Terra yoki Aqua
`confidence_level`	NASA bu haqiqiy yong‘in ekaniga qanchalik ishonadi	Low / Nominal / High
`fire_power_mw`	Issiqlik kuchi, megavattda	21.3

Qiziqtiradigan bir nechta fakt:

- 🌾 Eng ko‘p yong‘in **mart** oyida bo‘lgan (924 ta). Ko‘p fermerlar erta bahorda dalalarni yoqadi.
- 🏭 **4 649 ta nuqtadan 2 000 tasi** yong‘in emas, balki gaz mash’allari yoki sanoat.
- 🌙 Taxminan har **3 ta** kuzatuvdan biri **kechasi** bo‘lgan.

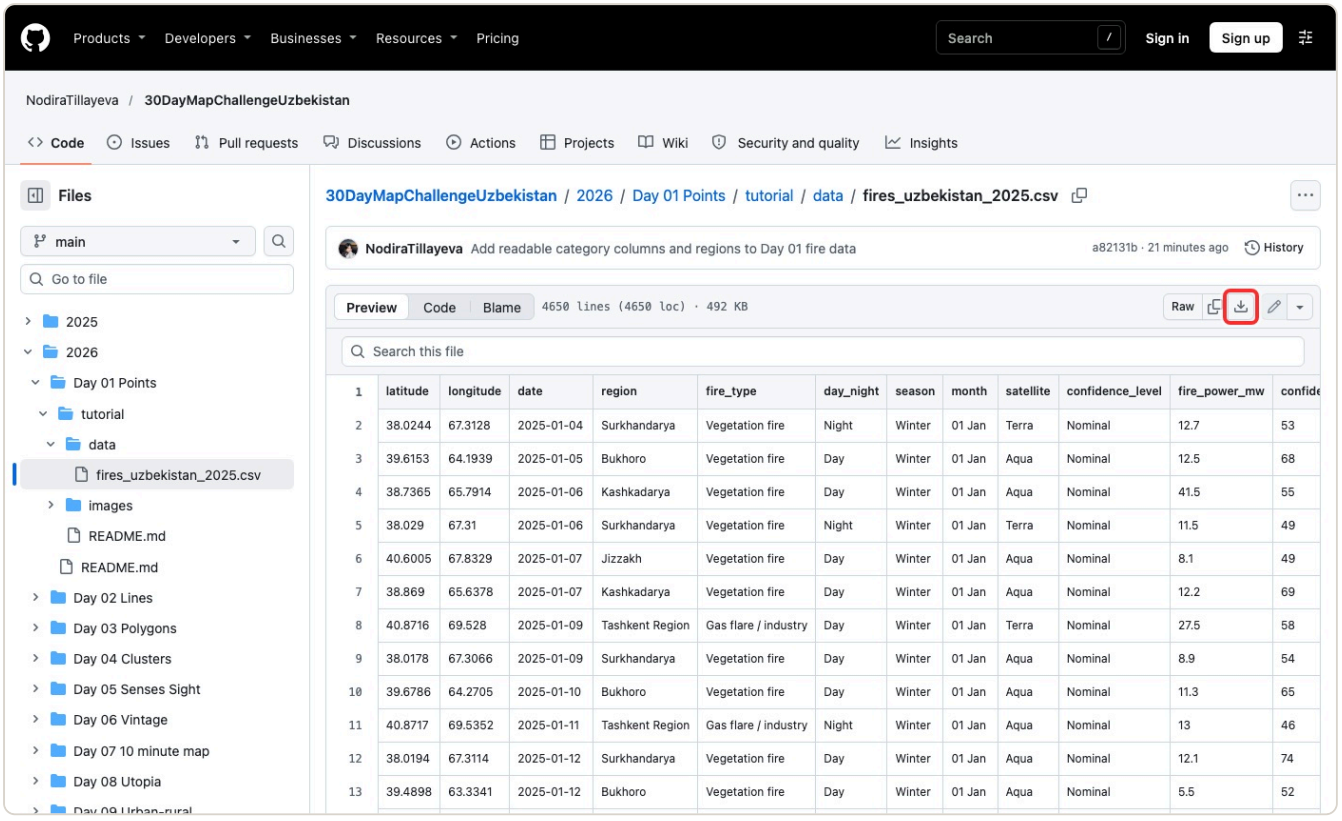
1-qism – Ma’lumotni yuklab oling

Avval ma’lumotni kompyuteringizga yuklab oling. Ikki yo‘l bor, ikkalasi ham bepul va ro‘yxatdan o‘tish shart emas.

	A variant – Bizning tayyor fayl	B variant – To‘g‘ridan-to‘g‘ri NASAdan
Kimga mos	Birinchi marta qilayotganlarga ✅	Boshqa yil yoki boshqa davlatni sinab ko‘rish uchun
Ustunlar	Oson so‘zlar: *Day/Night*, *Spring*, *Vegetation fire*, viloyat nomlari	NASAning qisqa kodlari: `D` / `N`, `0` / `2`, raqamlar

A variant – Tayyor faylimizni GitHubdan yuklab oling

1-qadam. **fires_uzbekistan_2025.csv** faylini oching. GitHub ma’lumotni jadval ko‘rinishida ko‘rsatadi. Jadval ustidagi o‘ng tomonda turgan **yuklab olish belgisini** ⬇️ bosing. `fires_uzbekistan_2025.csv` fayli **„Yuklanmalar“** (Downloads) papkasiga saqlanadi.



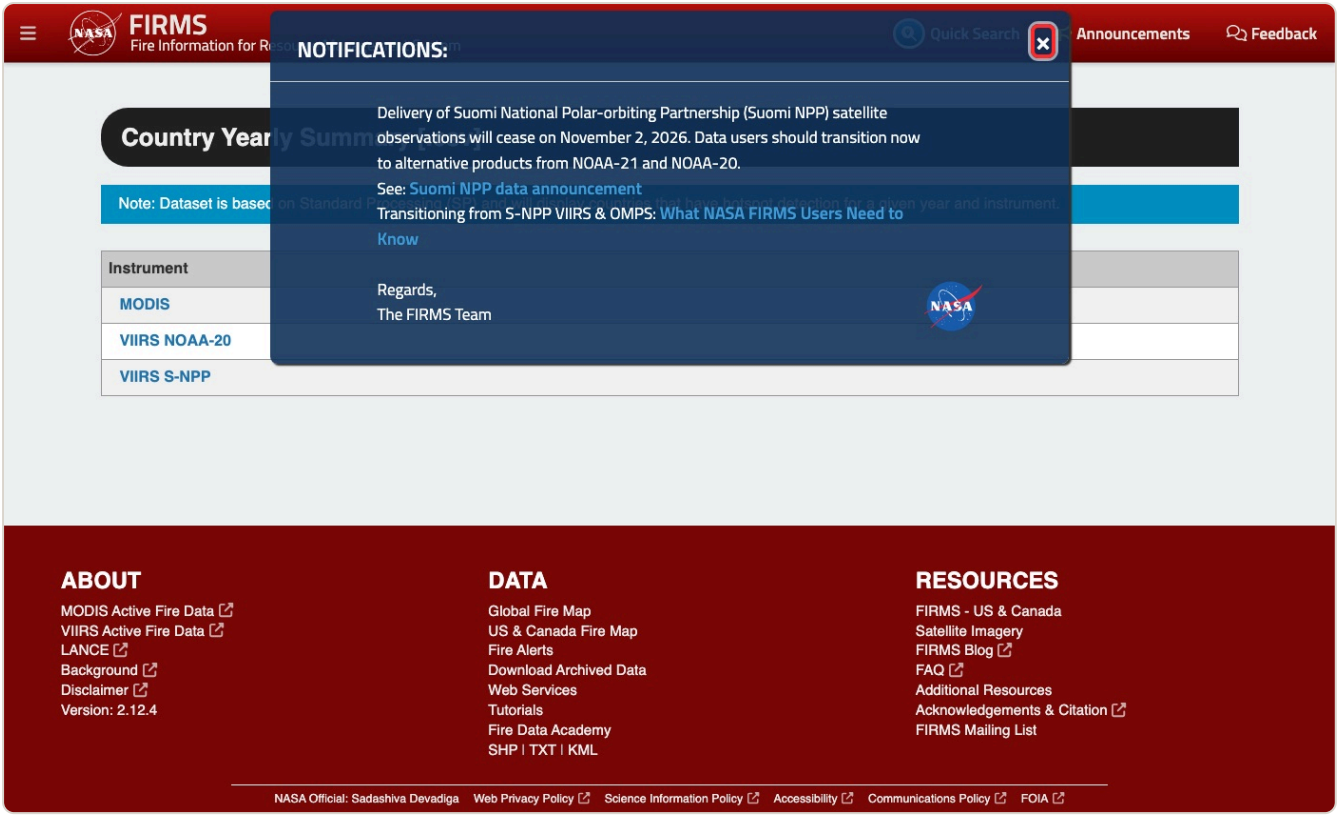
1	latitude	longitude	date	region	fire_type	day_night	season	month	satellite	confidence_level	fire_power_mw	confide
2	38.0244	67.3128	2025-01-04	Surkhandarya	Vegetation fire	Night	Winter	01 Jan	Terra	Nominal	12.7	53
3	39.6153	64.1939	2025-01-05	Bukhoro	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	12.5	68
4	38.7365	65.7914	2025-01-06	Kashkadarya	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	41.5	55
5	38.029	67.31	2025-01-06	Surkhandarya	Vegetation fire	Night	Winter	01 Jan	Terra	Nominal	11.5	49
6	40.6005	67.8329	2025-01-07	Jizzakh	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	8.1	49
7	38.869	65.6378	2025-01-07	Kashkadarya	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	12.2	69
8	40.8716	69.528	2025-01-09	Tashkent Region	Gas flare / industry	Day	Winter	01 Jan	Terra	Nominal	27.5	58
9	38.0178	67.3066	2025-01-09	Surkhandarya	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	8.9	54
10	39.6786	64.2705	2025-01-10	Bukhoro	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	11.3	65
11	40.8717	69.5352	2025-01-11	Tashkent Region	Gas flare / industry	Night	Winter	01 Jan	Aqua	Nominal	13	46
12	38.0194	67.3114	2025-01-12	Surkhandarya	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	12.1	74
13	39.4898	63.3341	2025-01-12	Bukhoro	Vegetation fire	Day	Winter	01 Jan	Aqua	Nominal	5.5	52

➡️ Endi **2-qism**ga o‘ting.

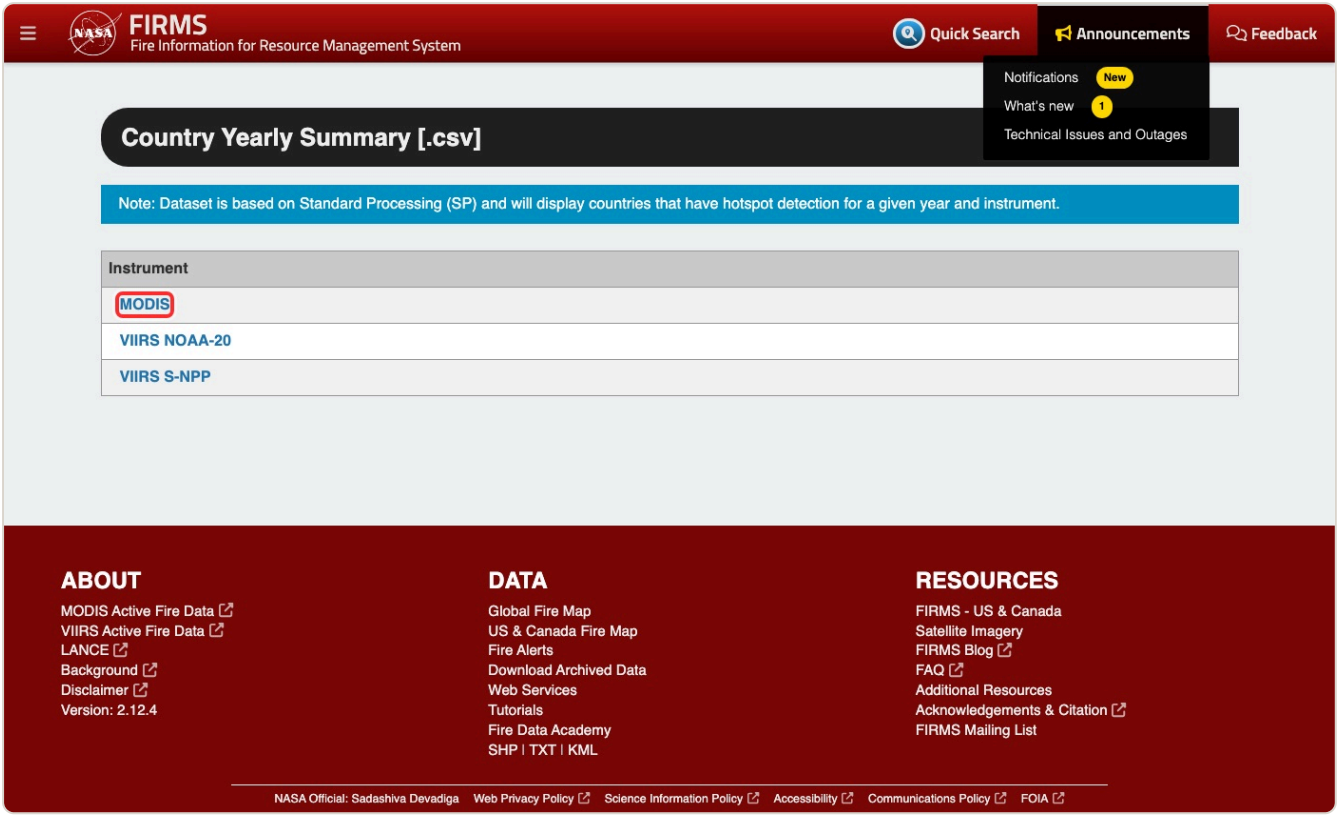
B variant – To‘g‘ridan-to‘g‘ri NASA FIRMS saytidan yuklab oling

Ma’lumot aslida shu yerdan olinadi. Bu to‘rtta bosishni bilib olsangiz, **istalgan davlat uchun, 2000-yildan bugungacha** yong‘in ma’lumotlarini olishingiz mumkin.

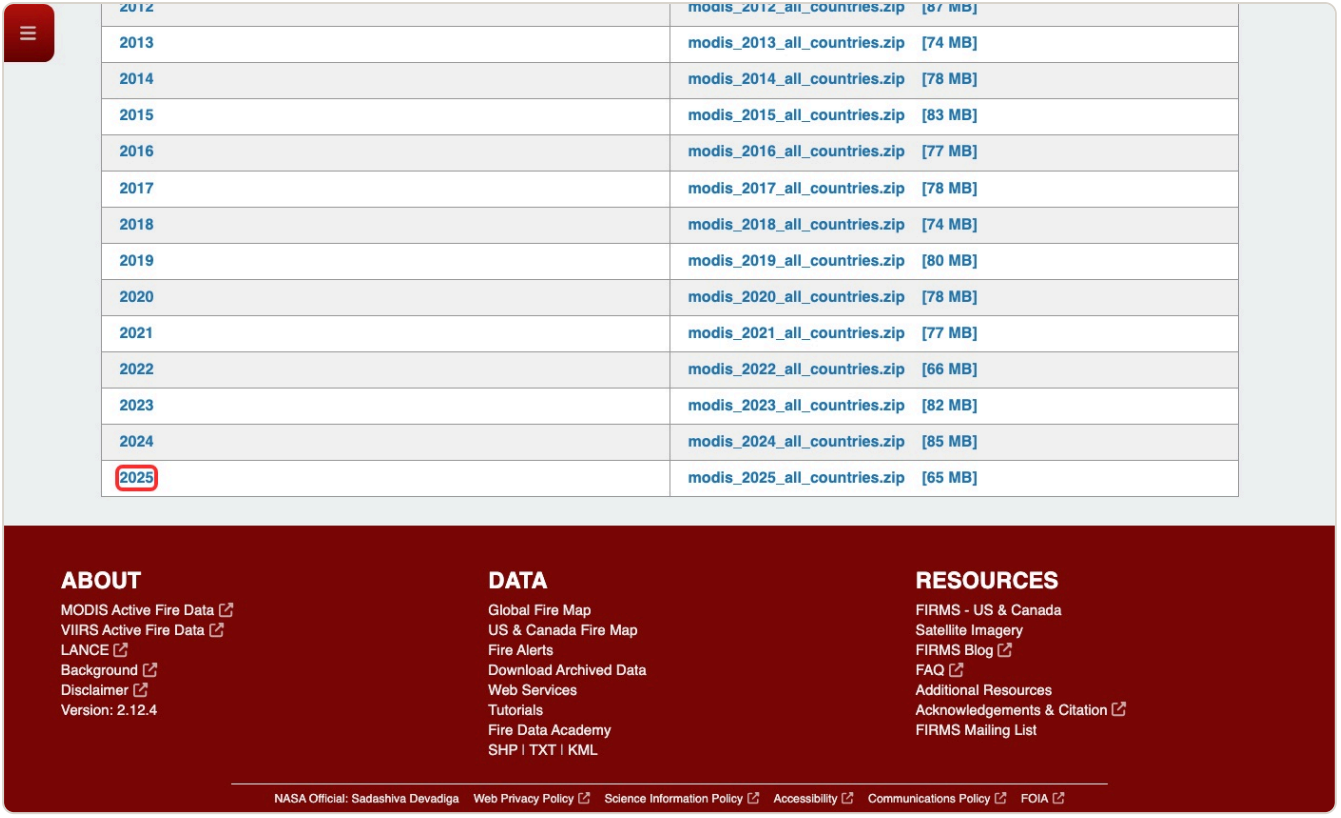
1a-qadam. firms.modaps.eosdis.nasa.gov/country saytini oching. Agar ko‘k rangli **NOTIFICATIONS** oynasi chiqsa, uni × belgisi bilan yoping.



1b-qadam. Instrument bo‘limida **MODIS** ni bosing. Bu Terra va Aqua sun’iy yo‘ldoshlaridagi asbob.

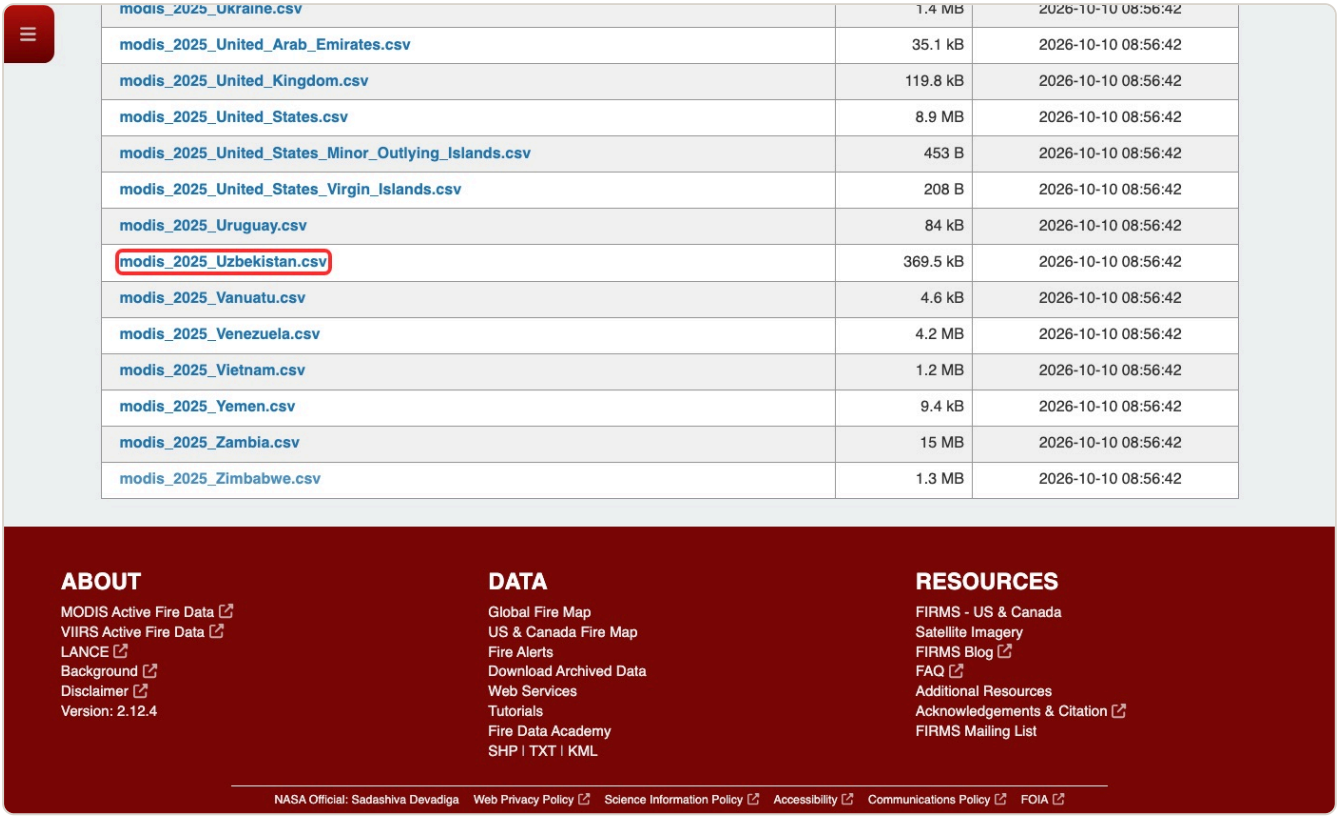


1c-qadam. Pastga tushib, kerakli **yilni** bosing, masalan **2025**. **Chap tomondagi** yil raqamini bosing, katta `.zip` faylni emas. Zip faylda dunyoning barcha davlatlari bor va u juda katta.



2012	modis_2012_all_countries.zip [87 MB]
2013	modis_2013_all_countries.zip [74 MB]
2014	modis_2014_all_countries.zip [78 MB]
2015	modis_2015_all_countries.zip [83 MB]
2016	modis_2016_all_countries.zip [77 MB]
2017	modis_2017_all_countries.zip [78 MB]
2018	modis_2018_all_countries.zip [74 MB]
2019	modis_2019_all_countries.zip [80 MB]
2020	modis_2020_all_countries.zip [78 MB]
2021	modis_2021_all_countries.zip [77 MB]
2022	modis_2022_all_countries.zip [66 MB]
2023	modis_2023_all_countries.zip [82 MB]
2024	modis_2024_all_countries.zip [85 MB]
2025	modis_2025_all_countries.zip [65 MB]

1d-qadam. Pastga tushib, davlatingizni toping va bosing, masalan **modis_2025_Uzbekistan.csv** (taxminan 370 kB). Fayl **„Yuklanmalar“** papkasiga saqlanadi.



modis_2025_Ukraine.csv	1.4 MB	2026-10-10 08:56:42
modis_2025_United_Arab_Emirates.csv	35.1 kB	2026-10-10 08:56:42
modis_2025_United_Kingdom.csv	119.8 kB	2026-10-10 08:56:42
modis_2025_United_States.csv	8.9 MB	2026-10-10 08:56:42
modis_2025_United_States_Minor_Outlying_Islands.csv	453 B	2026-10-10 08:56:42
modis_2025_United_States_Virgin_Islands.csv	208 B	2026-10-10 08:56:42
modis_2025_Uruguay.csv	84 kB	2026-10-10 08:56:42
modis_2025_Uzbekistan.csv	369.5 kB	2026-10-10 08:56:42
modis_2025_Vanuatu.csv	4.6 kB	2026-10-10 08:56:42
modis_2025_Venezuela.csv	4.2 MB	2026-10-10 08:56:42
modis_2025_Vietnam.csv	1.2 MB	2026-10-10 08:56:42
modis_2025_Yemen.csv	9.4 kB	2026-10-10 08:56:42
modis_2025_Zambia.csv	15 MB	2026-10-10 08:56:42
modis_2025_Zimbabwe.csv	1.3 MB	2026-10-10 08:56:42

> 💡 **Maslahat:** uzun ro‘yxatda davlatni tez topish uchun **Ctrl + F** (Mac’da **Cmd + F**) ni bosing va davlat nomini yozing.

📖 NASA ustunlarini qanday o‘qish kerak (faqat B variant uchun)

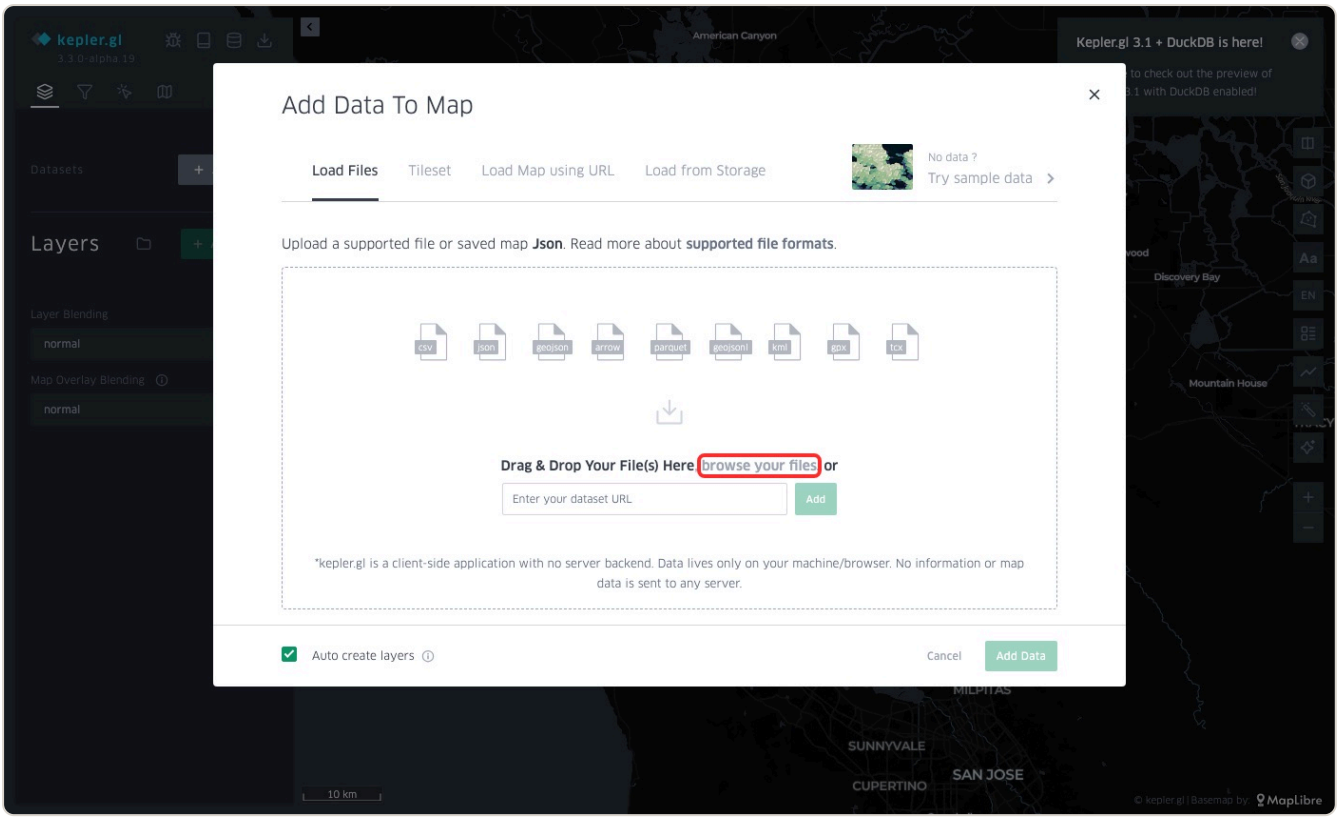
NASA faylida qisqa kodlar ishlatiladi. Ularning ma’nosi va tayyor faylimizdagi qaysi oson ustunga mos kelishi:

NASA ustuni	Ma’nosi	Bizning faylda
`acq_date`	Aniqlangan sana	`date`
`acq_time`	**UTC** bo‘yicha vaqt, SSDD ko‘rinishida (O‘zbekiston vaqti uchun 5 soat qo‘shing)	`acq_time_utc`
`daynight`	`D` = kunduz, `N` = tun	`day_night`
`type`	`0` = o‘simlik yong‘ini, `1` = vulqon, `2` = boshqa doimiy issiqlik manbai (gaz mash’allari, zavodlar), `3` = dengizda	`fire_type`
`confidence`	0–100, NASA bu haqiqiy yong‘in ekaniga qanchalik ishonadi	`confidence_level` (Low < 30, Nominal 30–80, High > 80)
`frp`	Fire Radiative Power: issiqlik kuchi, megavattda	`fire_power_mw`
`satellite`	`Terra` yoki `Aqua`	`satellite`
`brightness`	Piksel harorati, kelvinda	`brightness_k`

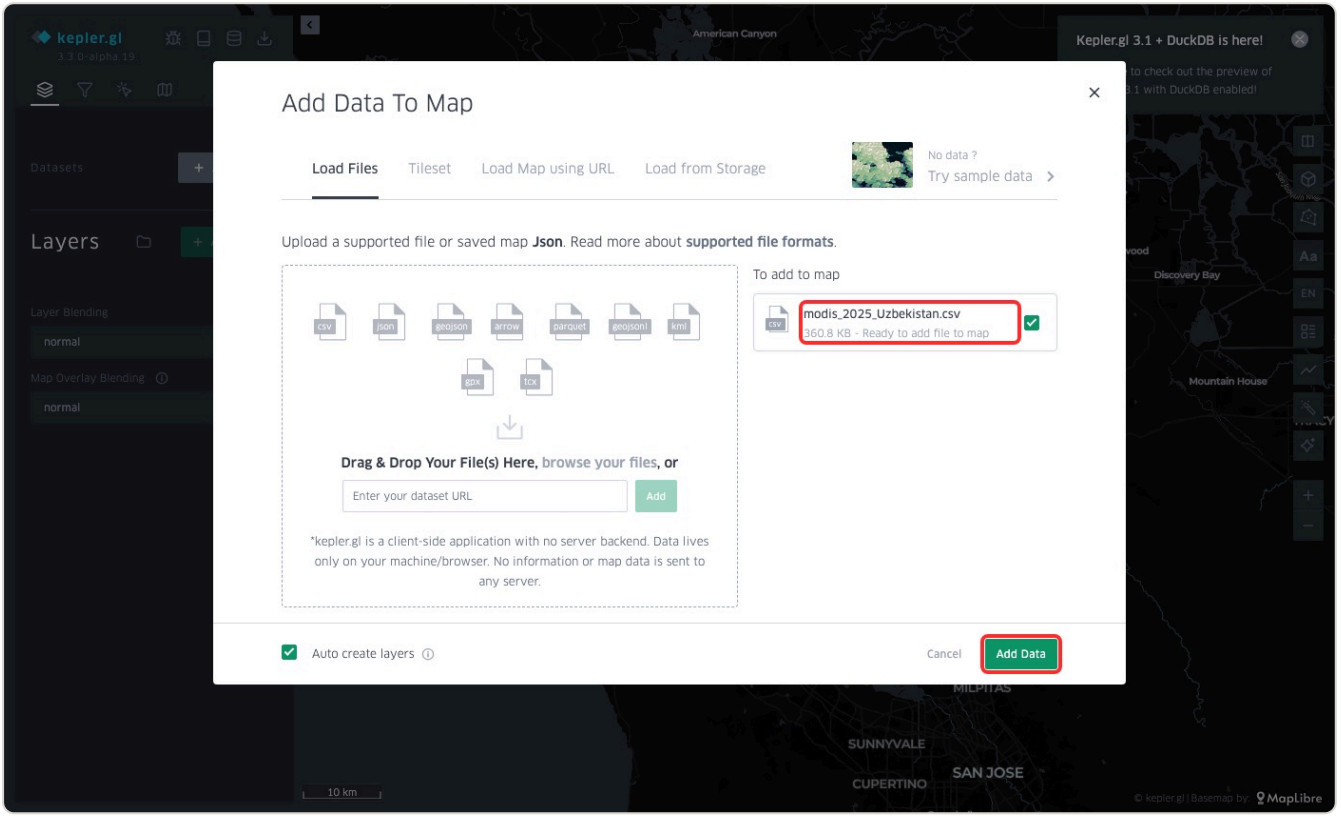
NASA faylida `region`, `season` va `month` ustunlari **yo‘q**. Ularni biz qo‘shganmiz. NASA fayli bilan ranglash uchun eng oson ustunlar **`daynight`** va **`satellite`**. `confidence` va `frp` esa ranglarning silliq o‘tishini beradi.

2-qism – Ma’lumotni kepler.gl ga yuklang

2-qadam. kepler.gl/demo saytini oching. **„Add Data To Map“** oynasi o‘zi ochiladi. **Yuklab olgan faylingizni** „Yuklanmalar“ papkasidan nuqtali ramka ichiga **sudrab tashlang** yoki **browse your files** ni bosib, faylni tanlang.



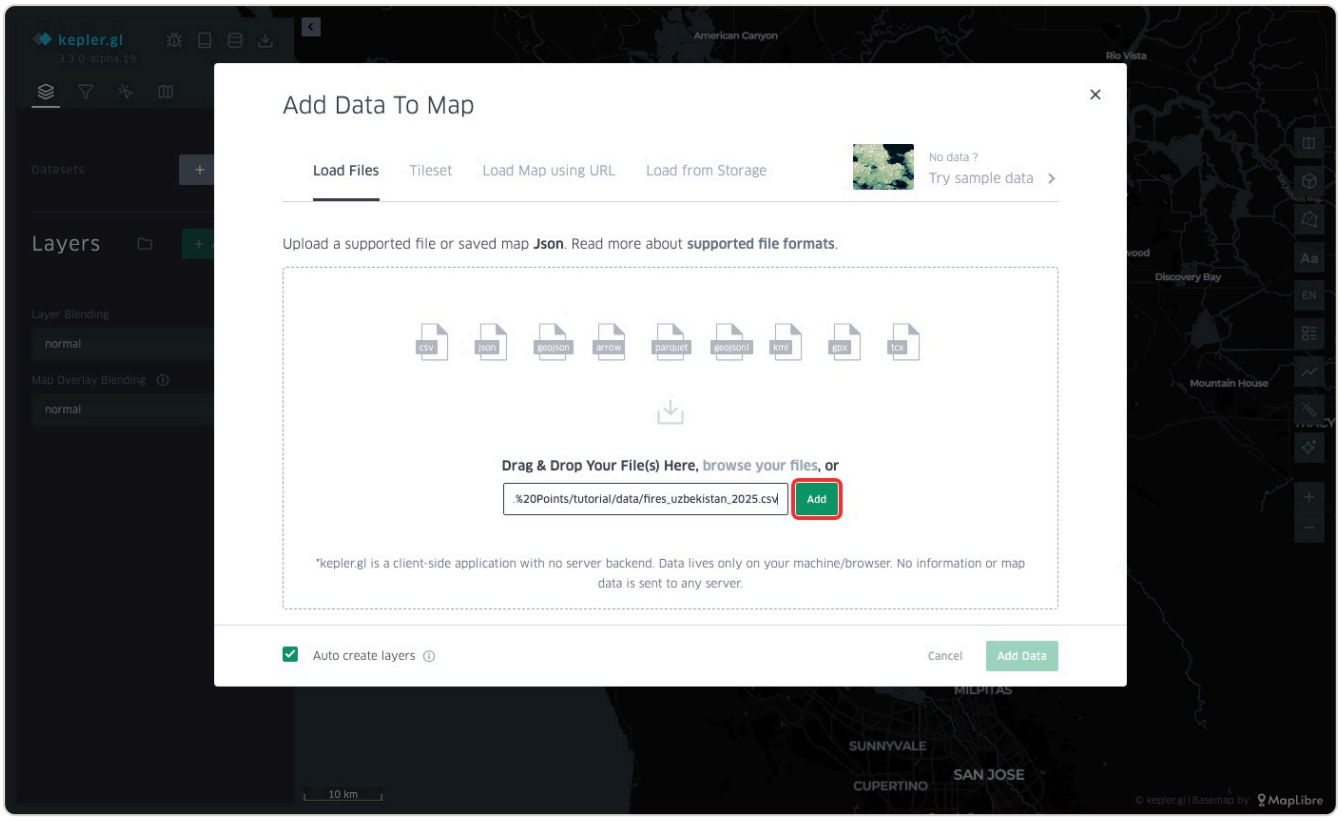
3-qadam. Fayl o‘ng tomonda paydo bo‘ladi. Yashil **Add Data** tugmasini bosing.



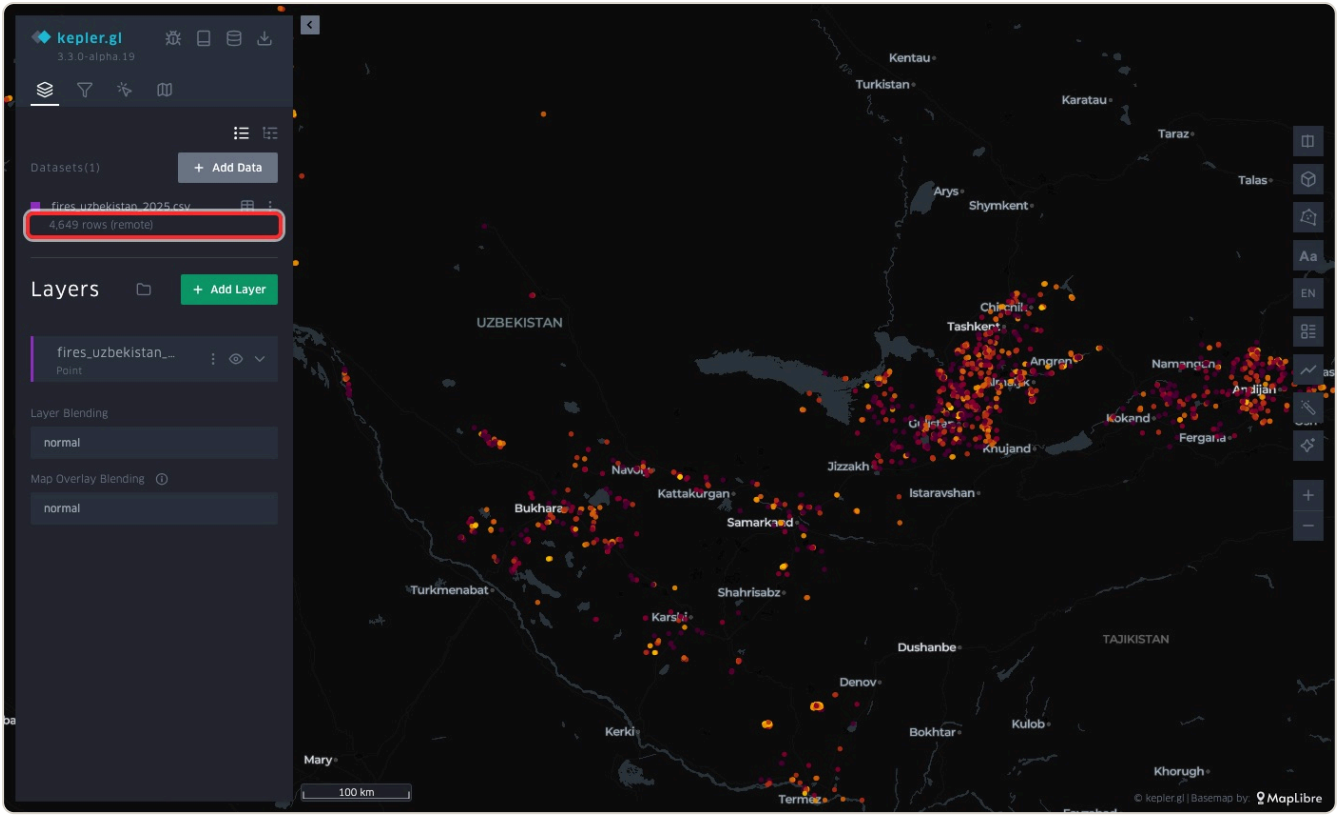
⚡ Tez yo‘l: faylimizni yuklab olmasdan yuklash (havolani qo‘yish)

Fayl o‘rniga bu havolani **„Enter your dataset URL“** maydoniga qo‘yib, **Add**, keyin **Add Data** ni bosishingiz mumkin:

```
https://raw.githubusercontent.com/NodiraTillayeva/30DayMapChallengeUzbekistan/main/2026/Day%2001%20Points/tutorial/data/fires_uzbekistan_2025.csv
```



4-qadam. 🎉 **Nuqtalaringiz xaritada!** Har bir nuqta kosmosdan ko‘ringan bitta yong‘in. Chap paneldagi **4,649 rows** yozuvi har bir nuqta uchun bitta qator borligini bildiradi.

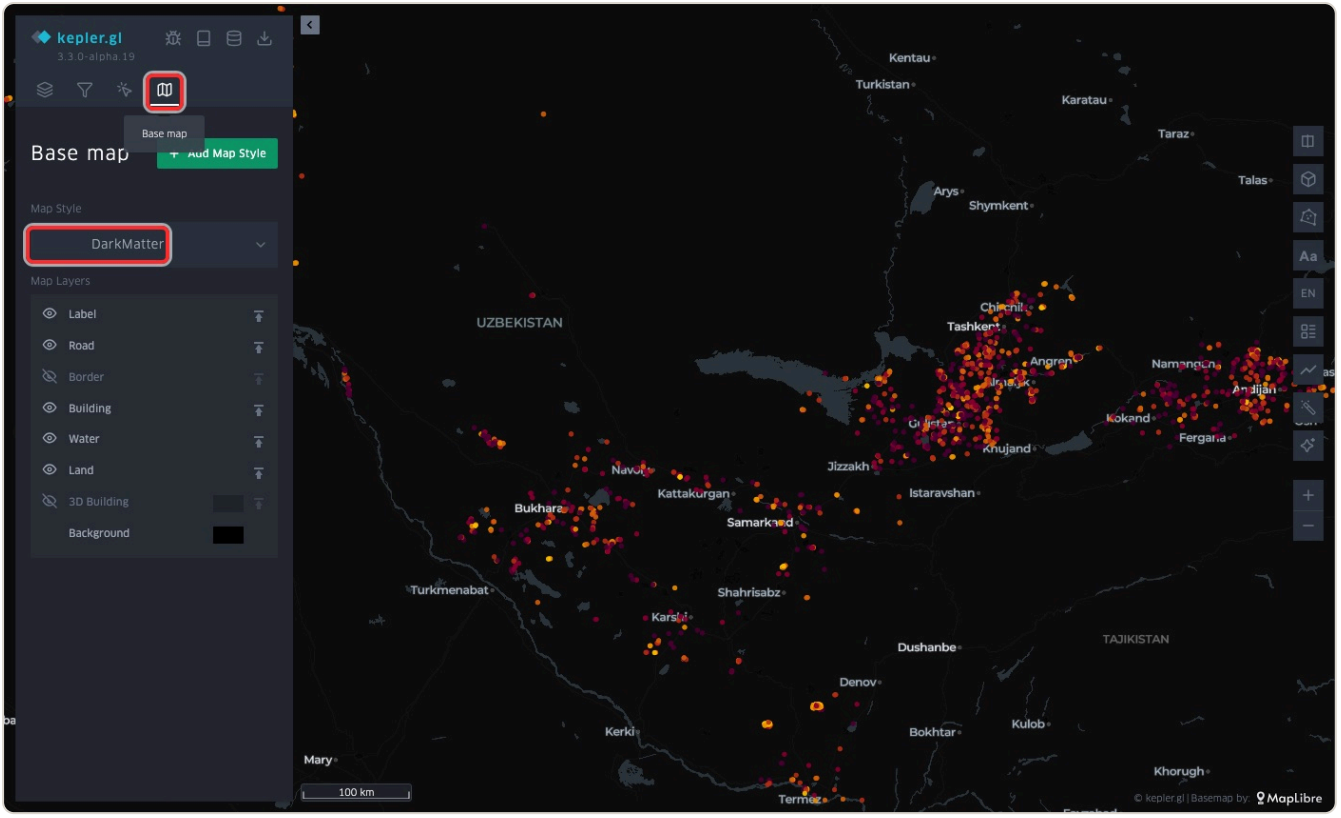


Xaritada harakatlanish: xaritani siljitish uchun sichqoncha bilan torting. Yaqinlashtirish va uzoqlashtirish uchun sichqoncha g‘ildiragini aylantiring. Xaritani qiyshaytirish yoki burish uchun sichqonchaning **o‘ng tugmasini** bosib turib torting.

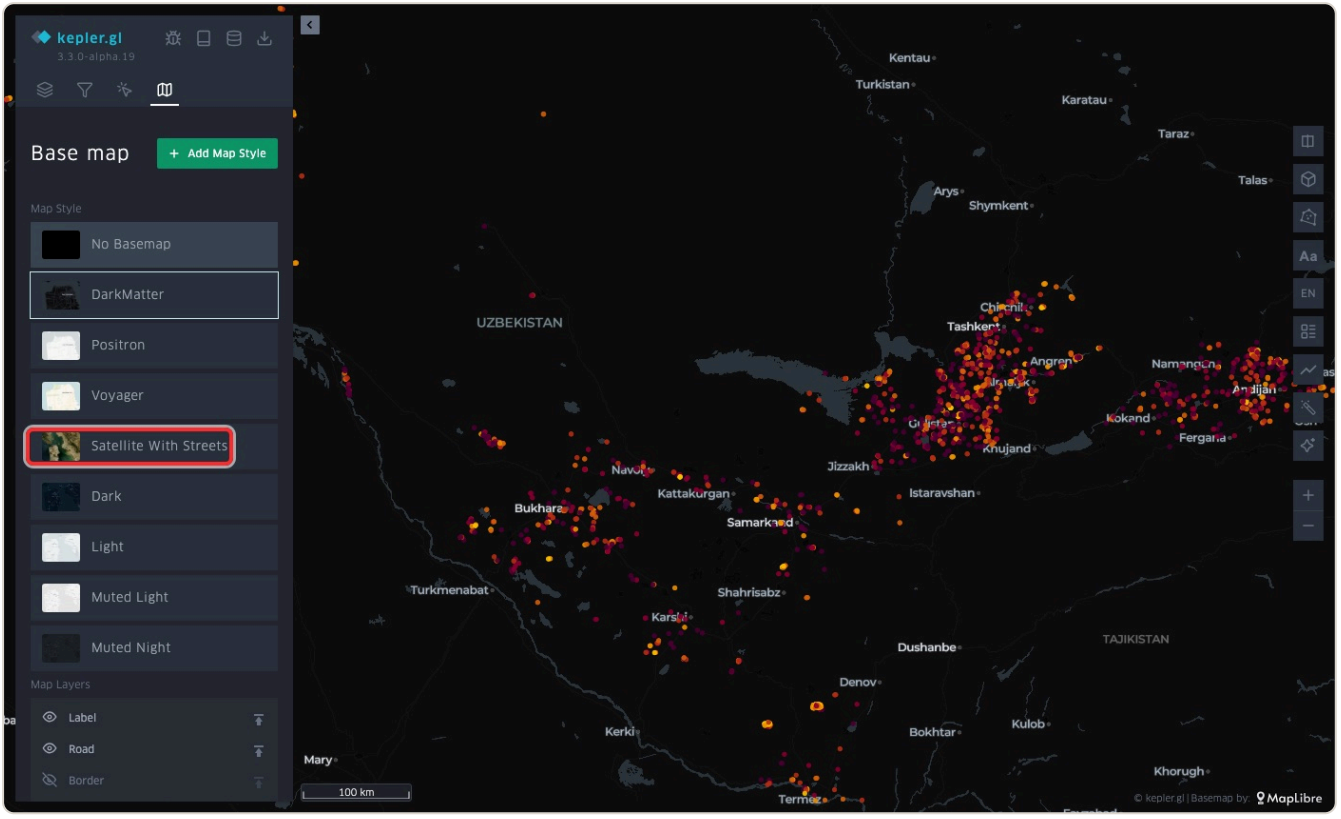
❌ Agar „DuckDB“ yoki so‘rovnoma haqida oyna chiqsa, uni × bilan yoping. Aks holda u yakuniy rasmingizga tushib qoladi.

3-qism – Nuqtalar ostiga sun’iy yo‘ldosh suratini qo‘ying

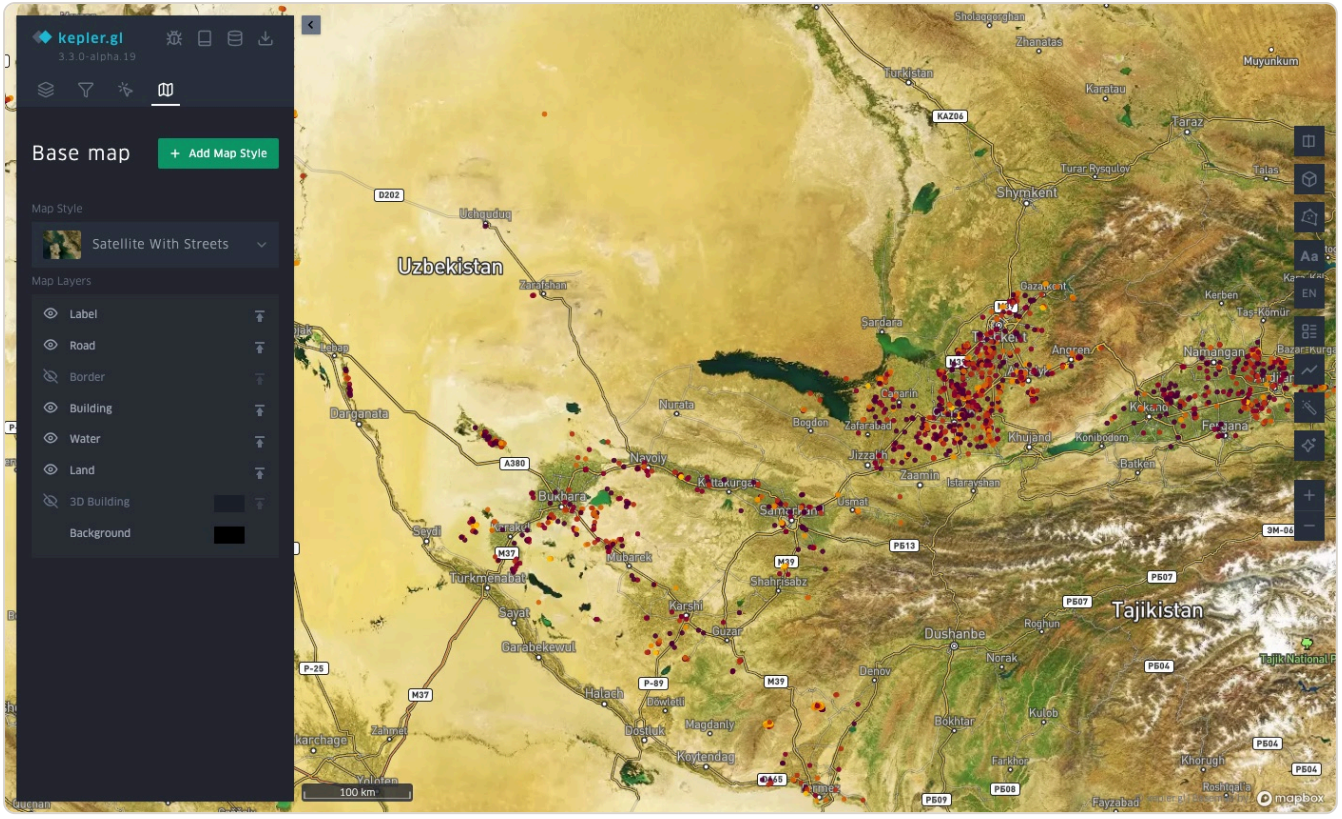
5-qadam. Xarita belgisini bosing (chap paneldagi 4-yorliq). Keyin *DarkMatter* deb yozilgan **Map Style** maydonini bosing.



6-qadam. Satellite With Streets ni tanlang.



7-qadam. Endi nuqtalaringiz ostida O‘zbekistonning cho‘llari, tog‘lari, daryolari va yashil dalalari ko‘rinadi. E’tibor bering: yong‘inlarning aksariyati cho‘lda emas, **yashil dalalarda**.

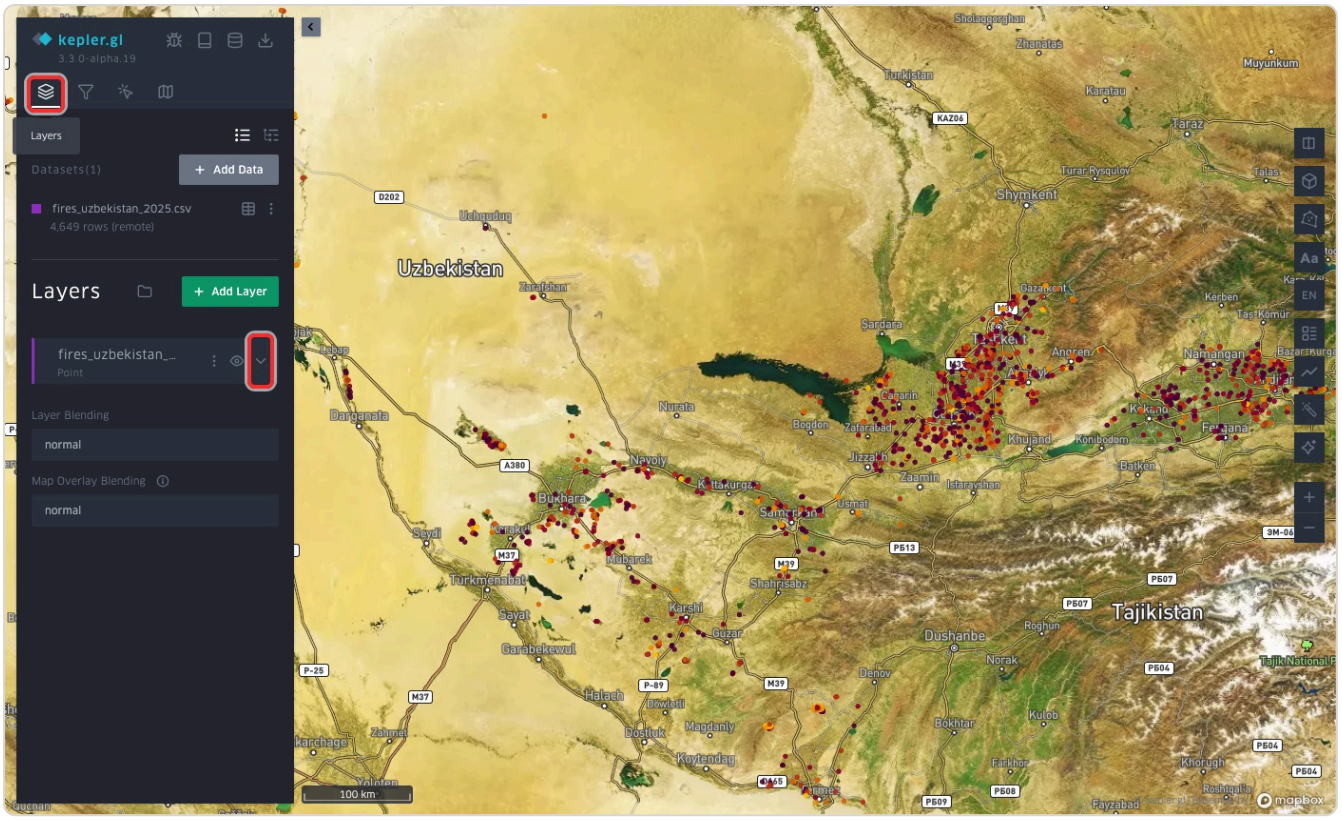


🎨 Sun’iy yo‘ldosh ko‘rinishi yoqmadimi? *Dark*, *Light* yoki *Voyager* ham chiroyli. Boshqa fon tanlash xaritani o‘zingizniki qilishning oson yo‘li.

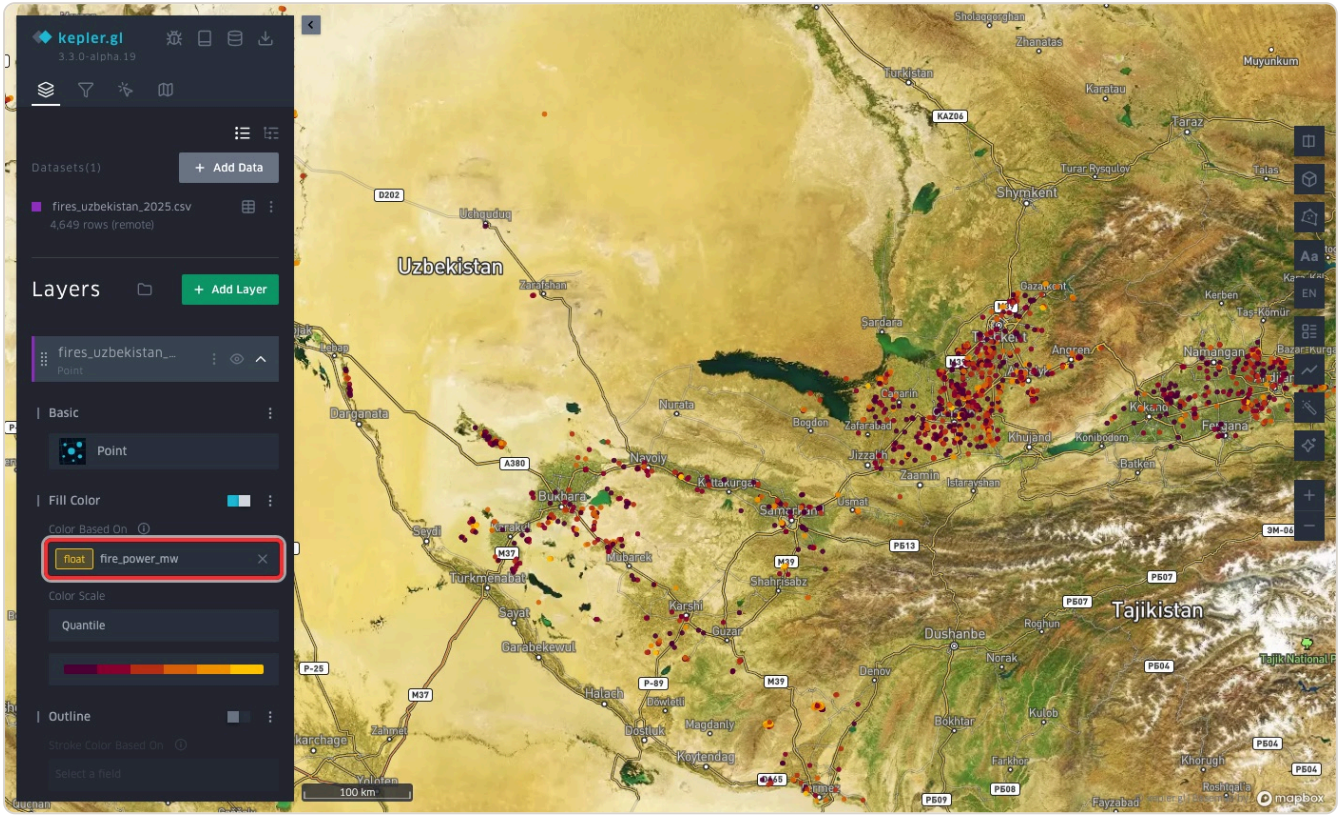
4-qism – Nuqtalarni toifalar bo‘yicha ranglang

Hozir nuqtalar yorqinlik bo‘yicha ranglangan. Keling, ularni ma’noliroq narsa bo‘yicha ranglaymiz.

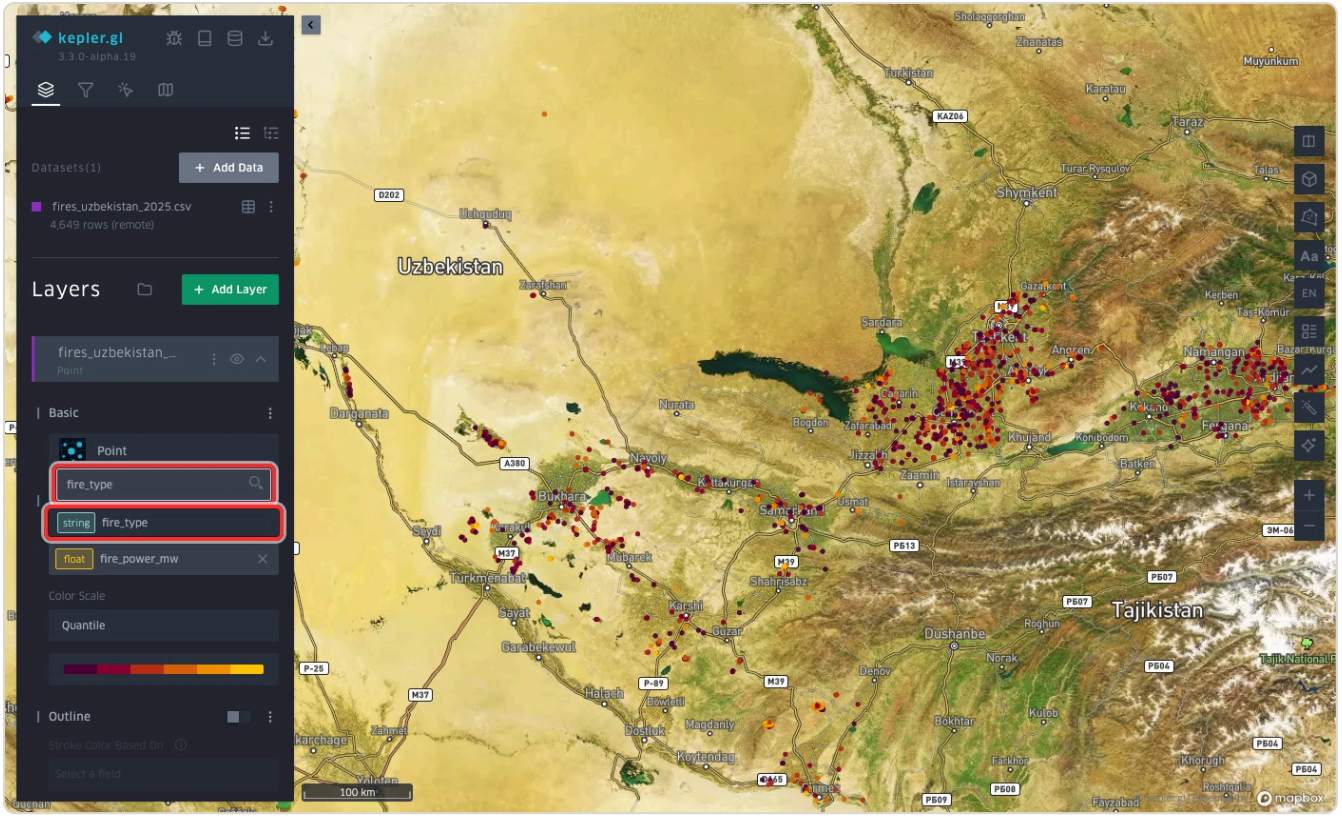
8-qadam. Qatlamlar belgisiga qayting (1-yorliq). Sozlamalarni ochish uchun qatlam nomi yonidagi kichik ⌄ **strelkani** bosing.



9-qadam. Fill Color bo‘limida **Color Based On** ostidagi maydonni bosing.

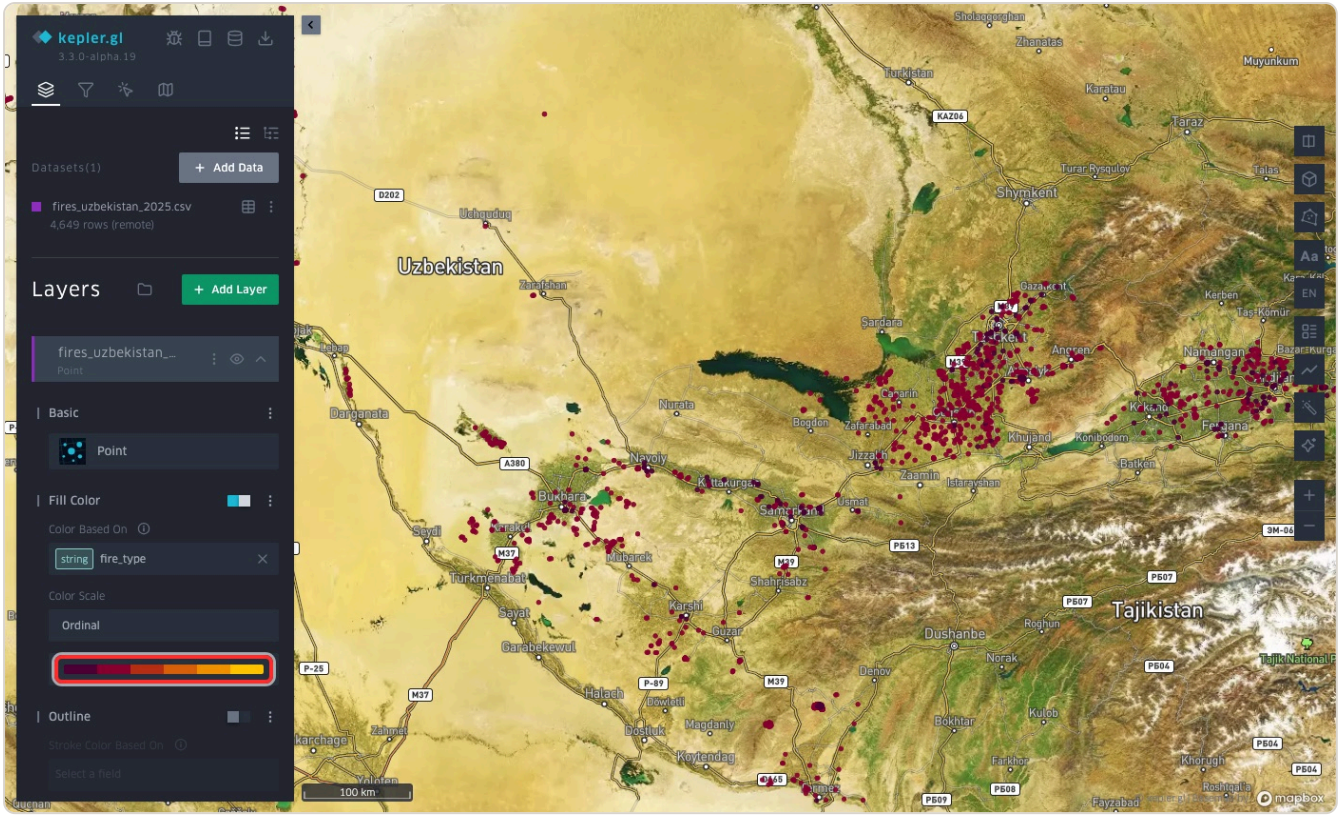


10-qadam. Qidiruv maydoniga ustun nomini yozing. Bu misolda `fire_type` dan foydalanamiz. Ro‘yxatdan uni bosing.

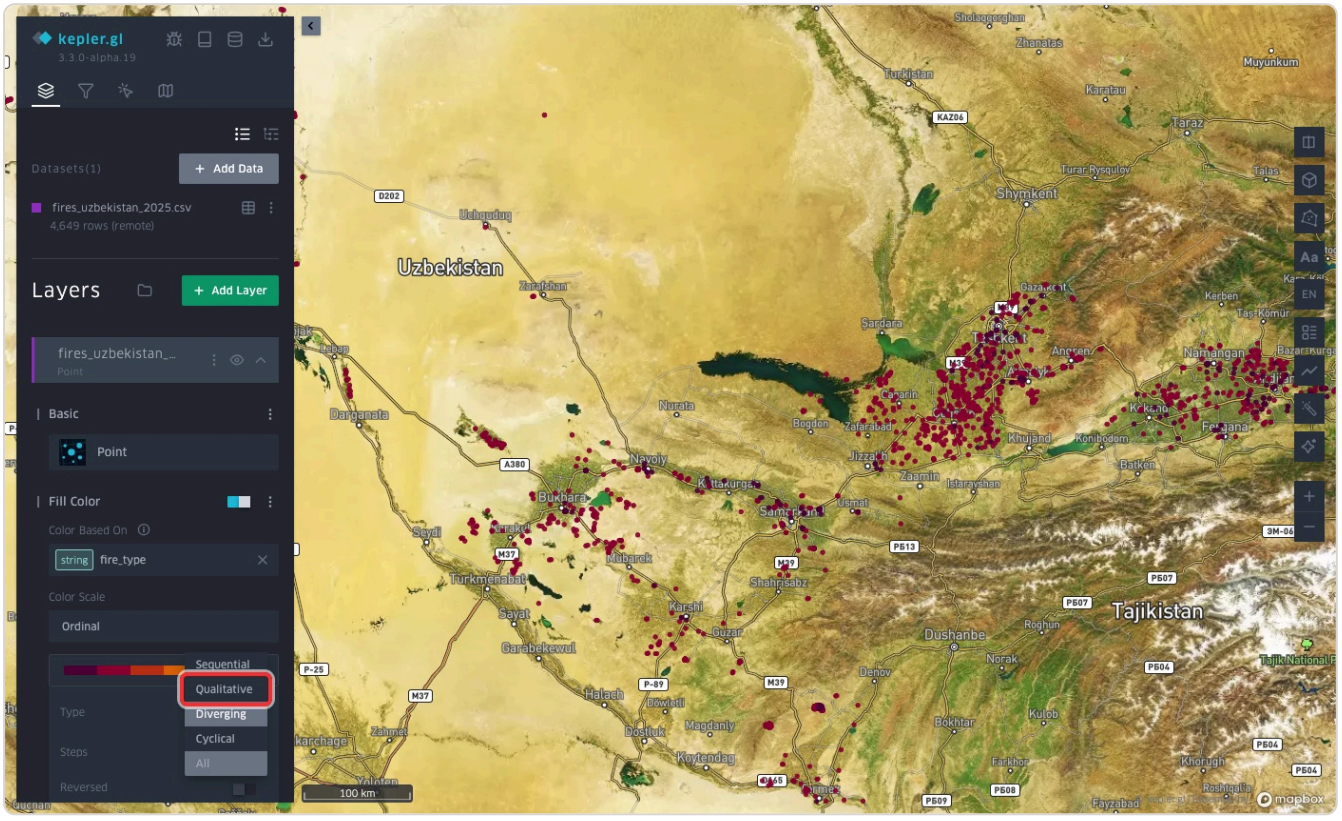


👉 **Aynan shu qadamda xaritangiz boshqalarnikidan farq qila boshlaydi. 6-qism**ga qarang va o‘z ustuningizni tanlang.

11-qadam. Ikki toifaning ranglari deyarli bir xil chiqishi mumkin. Buni tuzatish uchun *Color Scale* ostidagi **rangli chiziqni** bosing.



12-qadam. Type yonidagi **All** ni bosing va **Qualitative** ni tanlang. Qualitative palitralarda har bir toifa aniq farq qiladigan rangga ega bo‘ladi.



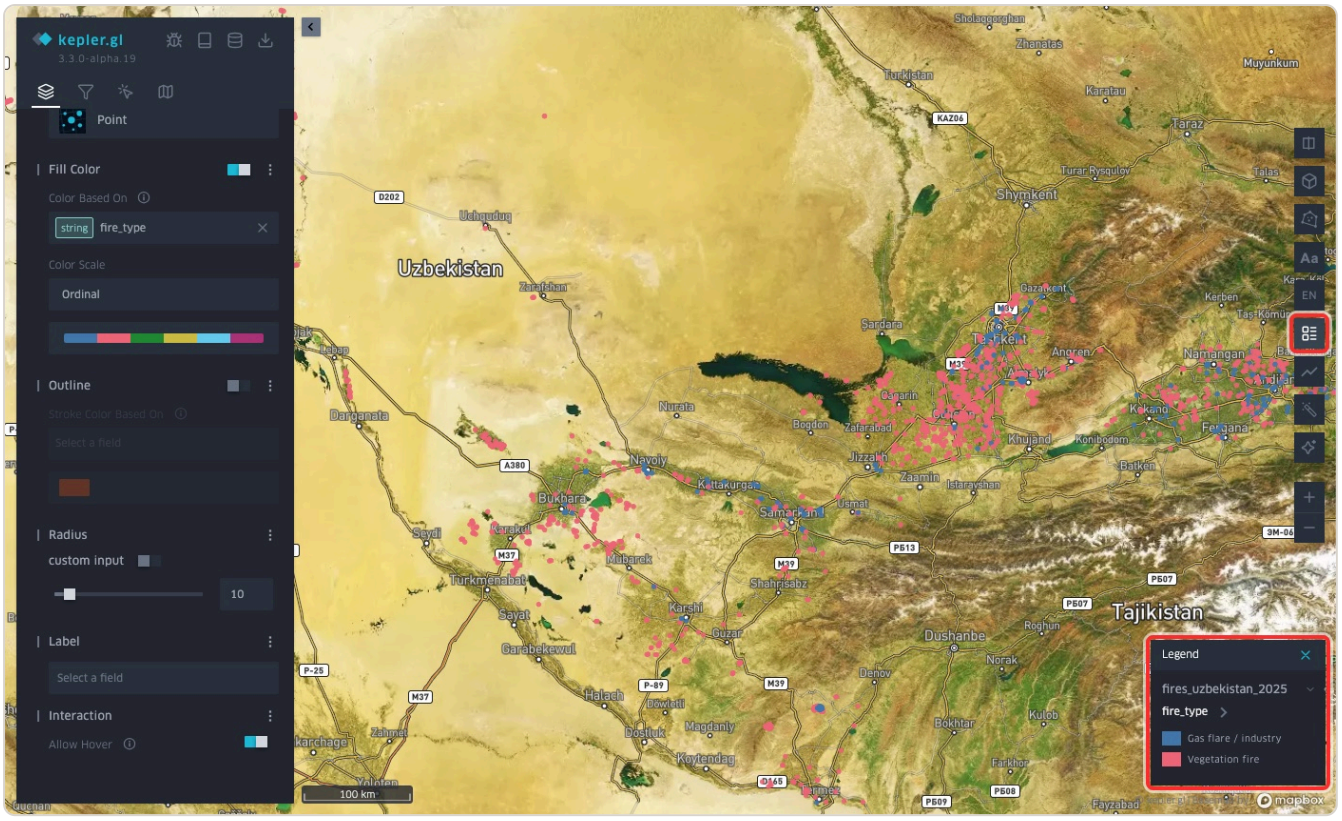
13-qadam. Ro‘yxatdan yoqqan palitrani bosing.



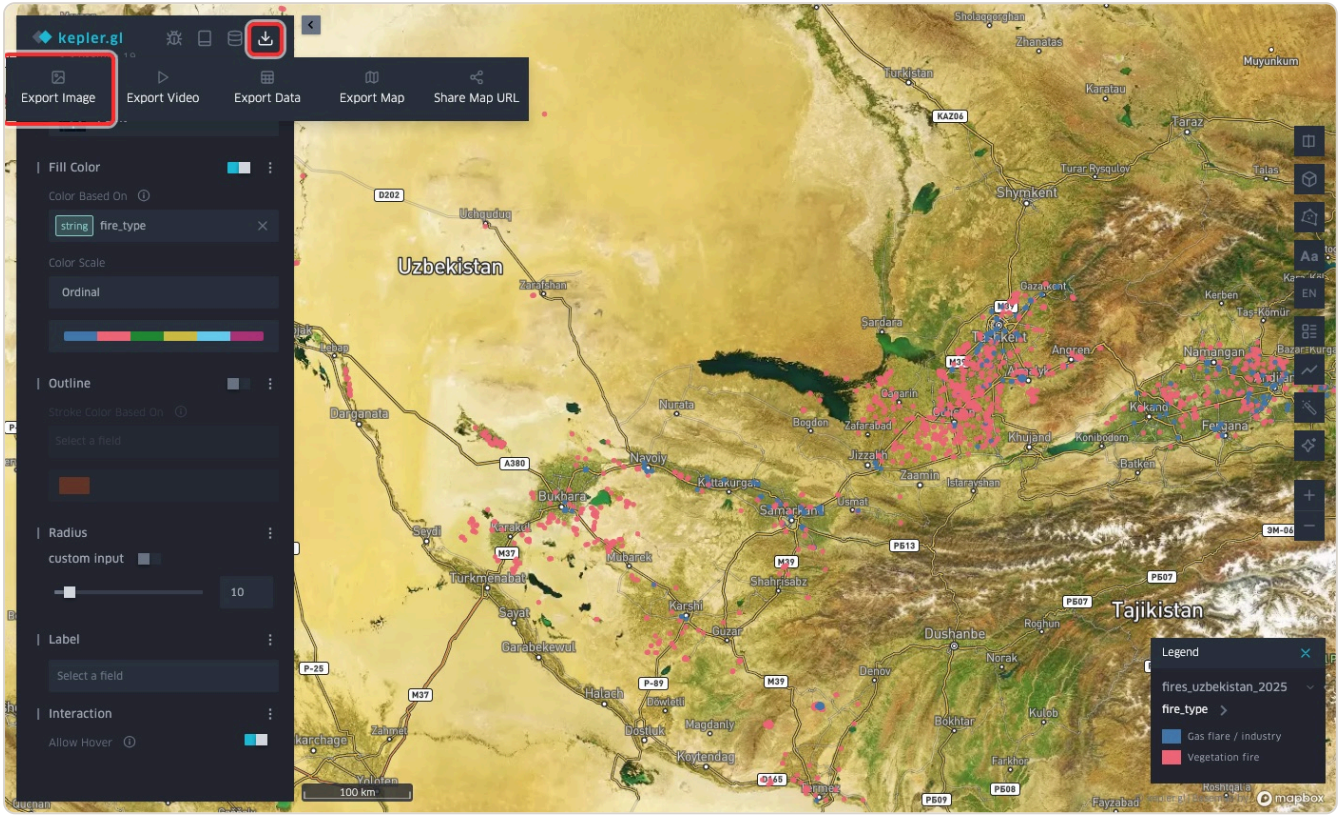
🔢 **Toifalar ko‘pmi?** (`region` da 14 ta!) Ranglar takrorlanmasligi uchun **Steps** ni 6 dan **12** ga o‘zgartiring.

5-qism – Shartli belgilarni qo‘shing va xaritani rasm qilib saqlang

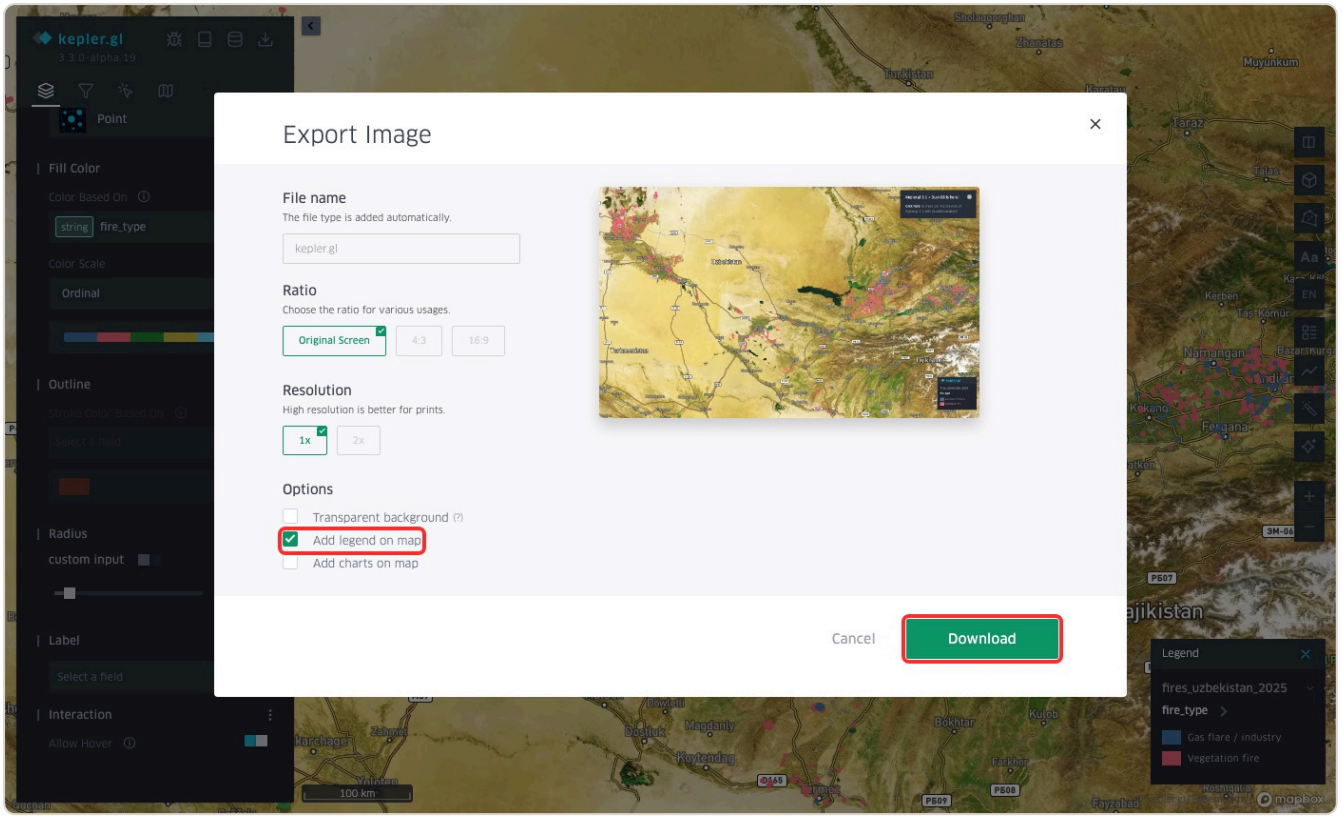
14-qadam. Xaritaning o‘ng tomonidagi **shartli belgilar (legend) tugmasini** bosing. Ranglarni tushuntiradigan kichik oyna paydo bo‘ladi. Endi *Gas flare / industry* va *Vegetation fire* aniq farq qiladigan ranglarda ko‘rinadi.



15-qadam. Xaritani o‘zingizga yoqadigan ko‘rinishga kelguncha yaqinlashtiring va siljiting. Keyin **yuklab olish belgisini** bosing (chap panel tepasida) va **Export Image** ni tanlang.



16-qadam. Add legend on map ga belgi qo‘ying. Rasm aniqroq bo‘lishini istasangiz, **Resolution** bo‘limida **2x** ni tanlang. Oldindan ko‘rish rasmi chiqishini kuting, keyin **Download** ni bosing. `.png` fayl **„Yuklanmalar“** papkasiga saqlanadi.



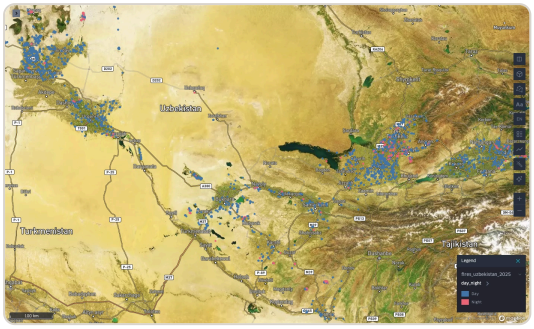
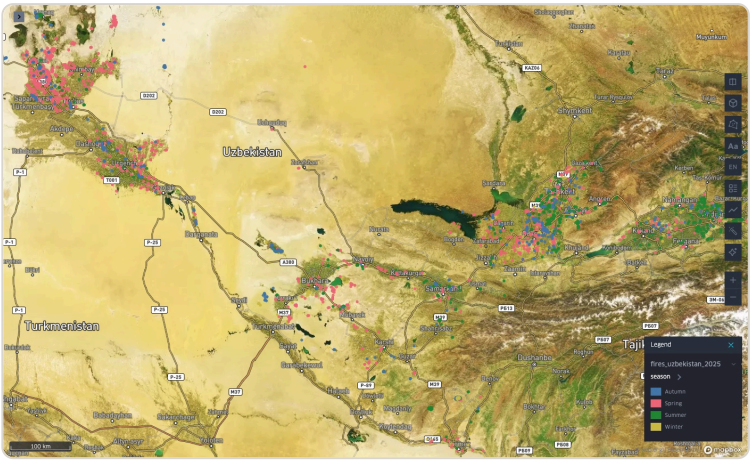
🎉 **Siz sun’iy yo‘ldosh ma’lumotlaridan xarita yaratdingiz!**

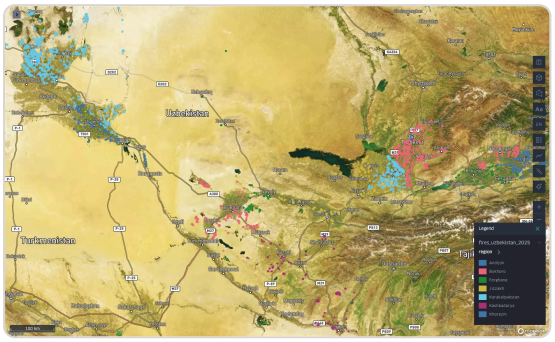
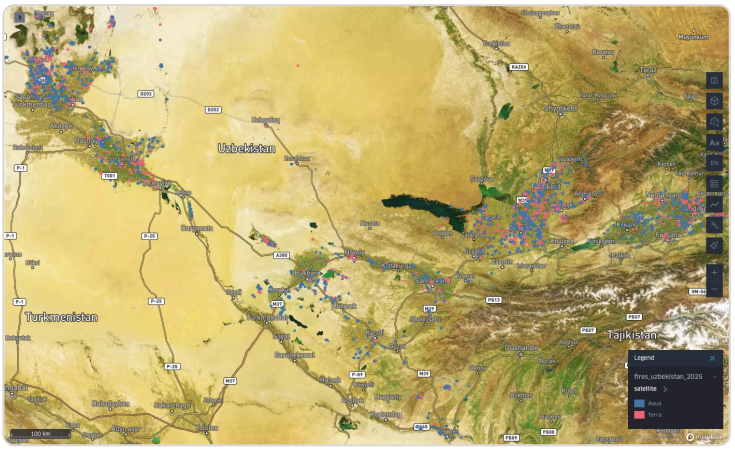
🎨 6-qism – Xaritani o‘zingizniki qiling

Guruhda hamma bir xil ma’lumotdan foydalanadi. **Sizning** xaritangizni boshqacha qilish yo‘li: **har bir qatordan bitta g‘oya** tanlang (yoki o‘zingiz o‘ylab toping!):

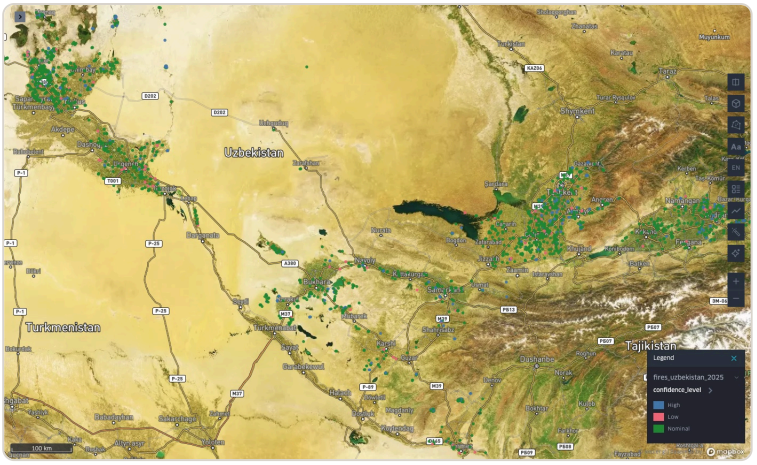
	Bittasini tanlang
1. Nima bo‘yicha ranglash (10-qadam)	`day_night` · `season` · `region` · `satellite` · `confidence_level` · `fire_type` · `month`
2. Qayerga qarash	Butun O‘zbekiston · o‘z viloyatingiz · Farg‘ona vodiysi · Qoraqalpog‘iston va Orol dengizi · bitta shahar
3. Fon (6-qadam)	Satellite · Dark · Light · Voyager · No basemap

Ba’zi variantlar qanday ko‘rinishi:

A · Kunduzmi yoki tunmi? (`day_night`)	**B · Qaysi fasl? (`season`)**
	
Kuzatuvlarning taxminan har uchtadan biri tunda bo‘lgan.	Xorazm va Qoraqalpog‘istonda Amudaryo bo‘yidagi dalalarda **bahorgi** yong‘inlar ustunlik qiladi.

C · Qaysi viloyat? (`region`)	**D · Qaysi sun’iy yo‘ldosh? (`satellite`)**
	
Har bir viloyat o‘z rangiga ega bo‘lishi uchun **Steps → 12** qiling.	Aqua (tushdan keyin o‘tadi) 2 679 ta nuqtani, Terra (ertalab o‘tadi) 1 970 tasini aniqlagan.

E · NASA qanchalik ishonadi? (`confidence_level`)



Ajralib turish uchun yana g‘oyalar ⭐

- **Kuchiga qarab o‘lcham:** qatlam sozlamalarida **Radius** ni oching va **Radius Based On** da `fire_power_mw` ni tanlang. Kuchli yong‘inlar katta doiralarga aylanadi.
- **Faqat bitta oy yoki fasl: filtr yorlig‘ini** oching (voronka belgisi, 2-yorliq) → **Add Filter** → `month` yoki `season` ni tanlang va kerakli qiymatlarni belgilang.
- **Yong‘inlar issiqlik xaritasi sifatida: Basic** bo‘limida *Point* ni **Heatmap** ga o‘zgartiring, zichlikni ko‘rsatuvchi yorqin xarita hosil bo‘ladi.
- 🕵️ **Topishmoqni yeching:** taxminan **1 700 ta issiq nuqta** Surxondaryoda, **38.05° sh. k., 67.30° sh. u.** yaqinida deyarli bitta joyga to‘plangan. Ular bitta nuqtaga o‘xshaydi! Sun’iy yo‘ldosh fonida o‘sha joyni yaqinlashtiring. Sizningcha, u yerda yil bo‘yi nima yonmoqda?
- **Yillarni solishtiring:** NASAdan boshqa yilni yuklab oling (**B variant**), masalan 2015 va 2025, va ikkala faylni ham yuklang. Har biri alohida qatlam bo‘ladi.

📤 7-qism – Xaritangizni ulashing

1. Xaritangiz **mavzusi** nomi bilan papka yarating, masalan `Fires by season`. Ichiga rasmingizni va **ismingiz**, ma’lumot va vositalar yozilgan qisqa `README.md` faylini qo‘ying.
2. Papkani butunligicha repozitoriyamizdagi **`2026/Day 01 Points/`** papkasiga yuklang. Kodsiz qanday qilish asosiy README dagi Quick Start bo‘limida yozilgan.
3. Xaritani **#30DayMapChallenge** heshtegi bilan joylang va nimalarni tanlaganingizni yozing (masalan, *„fasllar bo‘yicha ranglangan, Farg‘ona vodiysi, qora fon“*).

🆘 Nimadir noto‘g‘ri ketdimi?

Muammo	Yechim
Add Data oynasi yo‘qolib qoldi	Chap panel tepasidagi **+ Add Data** ni bosing.
Havolani qo‘ygandan keyin hech narsa bo‘lmayapti	Havolani **to‘liq** nusxalaganingizni tekshiring. Yoki faylni yuklab olib, sudrab tashlang (1-qism).
Yuklab olingan faylni topa olmayapman	**„Yuklanmalar“** papkasiga qarang. Chrome’da yuklanmalarni ko‘rish uchun **Ctrl + J** (Mac’da Cmd + Shift + J) ni bosing.
Barcha nuqtalar bir xil rangda	**Color Based On** da ustun tanlaganingizni tekshiring (10-qadam).
Ranglar juda o‘xshash	**Qualitative** palitradan foydalaning (11–13-qadamlar).
Xarita bo‘sh yoki kulrang	Fon yuklanishi uchun bir necha soniya kuting yoki xaritani bir marta yaqinlashtirib-uzoqlashtiring.
Yorliqni yopib qo‘ydim va hammasi yo‘qoldi	Internetda hech narsa saqlanmaydi. Faylni yana sudrab tashlang (2-qadam). Bu bir daqiqa oladi, xolos.

📚 Ma’lumot manbalari

- **Yong‘inlar:** NASA FIRMS, MODIS Collection 6.1, faol yong‘inlar va issiq nuqtalar (MCD14ML), Terra va Aqua, O‘zbekiston, 2025. firms.modaps.eosdis.nasa.gov. doi: 10.5067/FIRMS/MODIS/MCD14ML
- **Viloyat nomlari:** `Boundaries` papkasidagi viloyat chegaralari asosida qo‘shilgan (manba: simplemaps.com). Chegaradan sal tashqaridagi nuqtalar eng yaqin viloyatga biriktirilgan.
- **Tushunarli ustunlar** (`fire_type`, `day_night`, `season`, `month`, `confidence_level`) FIRMSning asl kodlaridan tayyorlangan: `type`, `daynight`, `acq_date` va `confidence` (Low < 30, Nominal 30–80, High > 80).